



Sydney Girls High School

2014

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

10 Marks

- Attempt Questions 1 – 10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II

90 Marks

- Attempt Questions 11 – 16
- Answer on the blank paper provided
- Begin a new page for each question
- Allow about 2 hours and 45 minutes for this section

Name:

Teacher:

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format or the content of the 2014 HSC Examination Paper in this subject.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. The substitution $t = \tan \frac{\theta}{2}$ is used to find $\int \sec \theta d\theta$. Which of the following gives the correct expression for the required integral?

(A) $\int \frac{dt}{2(1-t^2)}$

(B) $\int \frac{2tdt}{1-t^2}$

(C) $\int \frac{2dt}{1-t^2}$

(D) $\int \frac{4tdt}{(1+t^2)^2}$

2. The distance between the foci for the ellipse $\frac{x^2}{100} + \frac{y^2}{36} = 1$ is :

(A) 8

(B) 16

(C) 20

(D) 25

3. Which of the following is an expression for $\int \frac{dx}{\sqrt{7-6x-x^2}}$?

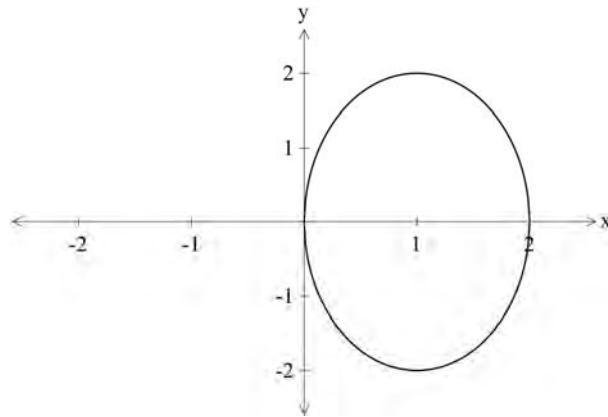
(A) $\sin^{-1}\left(\frac{x+3}{2}\right) + C$

(B) $\sin^{-1}\left(\frac{x+3}{4}\right) + C$

(C) $\sin^{-1}\left(\frac{x-3}{4}\right) + C$

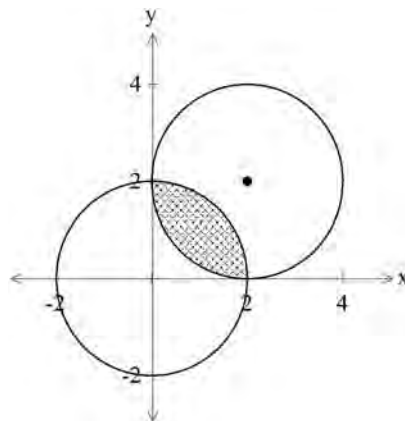
(D) $\sin^{-1}\left(\frac{x-3}{2}\right) + C$

4. The region enclosed by the ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y -axis to form a solid.



Which integral represents the volume of the solid when applying the method of slicing?

- (A) $V = \int_{-2}^2 \pi \sqrt{4-y^2} \, dy$
- (B) $V = \int_{-2}^2 2\pi \sqrt{1-y^2} \, dy$
- (C) $V = \int_{-2}^2 \pi \sqrt{1-y^2} \, dy$
- (D) $V = \int_{-2}^2 2\pi \sqrt{4-y^2} \, dy$
5. The shaded area in the Argand diagram below can be represented by the inequalities :



- (A) $|z| \leq 4$ and $|z - 2 + 2i| \leq 4$
- (B) $|z| \leq 2$ and $|z - 2 - 2i| \leq 2$
- (C) $|z| \leq 4$ and $|z - 2 - 2i| \leq 4$
- (D) $|z| \leq 2$ and $|z - 2 + 2i| \leq 2$

6. What is the solution to the inequation $\frac{x(5-x)}{x-4} \geq -3$?

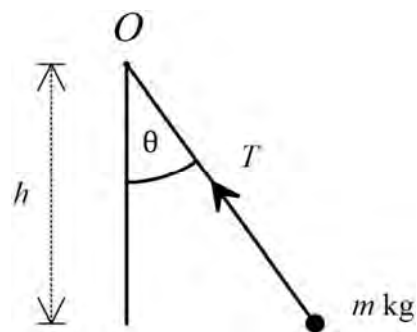
- (A) $2 \leq x < 4$ or $x \geq 6$
- (B) $4 < x \leq 6$ or $x \leq 2$
- (C) $4 > x \leq 5$ or $x \leq 1$
- (D) $1 \leq x < 4$ or $x \geq 5$

7. The gradient of the tangent at $(2,1)$ on the curve $2x^2 + 3xy + y^2 = 15$ is :

- (A) -1
- (B) $-\frac{8}{5}$
- (C) $-\frac{11}{2}$
- (D) $-\frac{11}{8}$

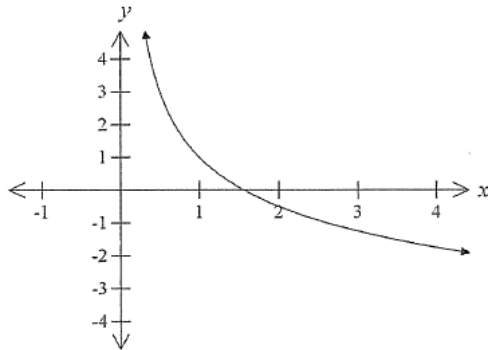
8. A particle of mass m moves in a horizontal circle with angular speed ω at a distance h below the point O . Which of the following reflects the relationship between ω and h ?

- (A) $h = \omega^2 g$
- (B) $h = \omega g$
- (C) $g = \omega^2 h$
- (D) $g = \omega h$

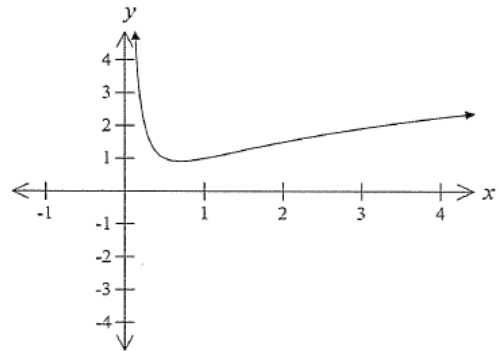


9. Which of the following is the graph of $y = \log_2 x + \frac{1}{x}$?

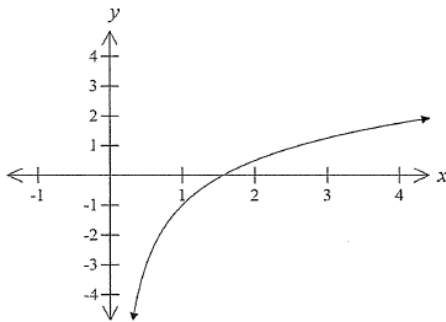
(A)



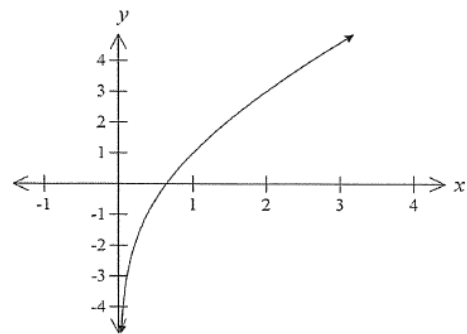
(B)



(C)



(D)



10. The polynomial $P(x) = x^4 + cx^2 + dx + 28$ has a double root at $x = 2$. What are the values of c and d ?

- (A) $c = -5$ and $d = -12$
 (B) $c = -5$ and $d = 12$
 (C) $c = -11$ and $d = 12$
 (D) $c = -11$ and $d = -12$

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Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer on the blank paper provided. Begin a new page for each question.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- (a) (i) Find numbers A , B and C such that 2

$$\frac{13}{(x+2)(x^2+9)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+9}.$$

- (ii) Hence find $\int \frac{13dx}{(x+2)(x^2+9)}.$ 2

- (b) (i) Express $z = \frac{1+2i}{1-3i}$ in modulus-argument form. 2

- (ii) Hence, find the value of z^7 , expressing your answer in the form $a+ib$, where a and b are real. 2

- (c) (i) Expand and simplify $(4+3i)^2$. 1

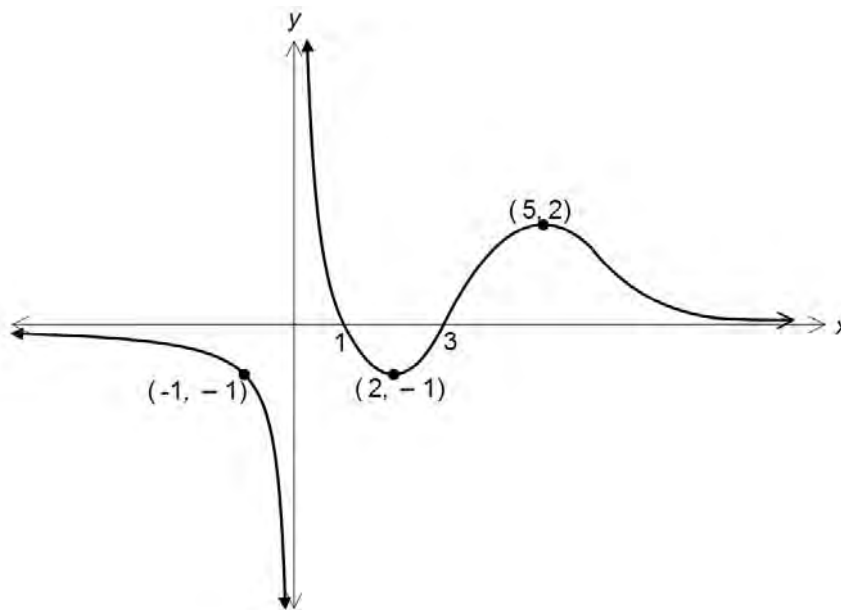
- (ii) Hence, solve the equation $2z^2 - iz - 1 - 3i = 0$, expressing the solutions in the form $x+iy$. 2

- (d) $(1+3i)$ is a root of the polynomial $P(x) = x^4 + 2x^3 + 11x^2 + 22x + 90$. 4

Find all the other roots.

Question 12 (15 marks)

- (a) The diagram shows the graph of $y = f(x)$.



Sketch the following curves on separate half-page diagrams :

- | | | |
|-------|----------------------|---|
| (i) | $y = f(-x)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y = f(x) - f(x) $ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |
- (b) Find $\int \frac{2x+1}{\sqrt{x^2-2x+10}} dx$.
- 3
- (c) $P\left(2p, -\frac{2}{p}\right)$ and $Q\left(2q, -\frac{2}{q}\right)$ are points on the rectangular hyperbola $xy = -4$.
- | | | |
|------|--|---|
| (i) | Show that the equation of the chord PQ is $x - pqy = 2p + 2q$. | 2 |
| (ii) | Find the locus of the mid-point of the chord given that the chord passes through the point $(0, -6)$. | 3 |

Question 13 (15 marks)

- (a) (i) If $I_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$, where n is a positive integer, show that

$$I_n = \frac{n}{n+1} I_{n-2}.$$

4

- (ii) Hence, evaluate I_5 . **2**

- (b) (i) By considering the expansion of $(\cos \theta + i \sin \theta)^5$, show that **3**
- $$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

- (ii) Hence, solve the equation $16x^4 - 20x^2 + 5 = 0$. Express the solutions in the form $x = \cos \alpha$. **3**

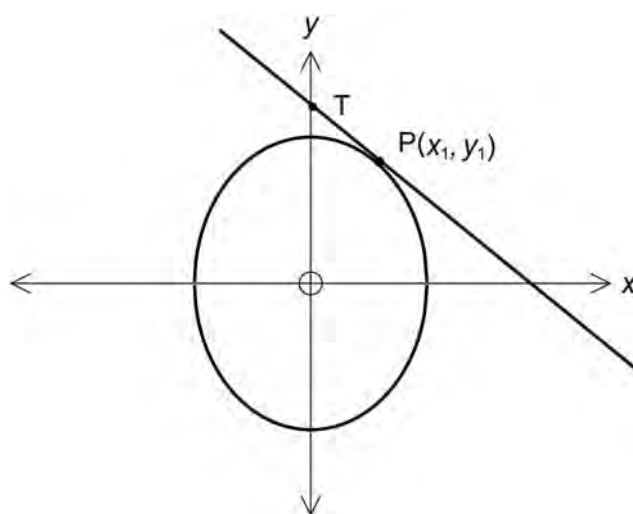
- (c) A string of length 40 cm can just sustain a weight of mass 12 kg without breaking. One end of the string is fixed to a point P on a smooth horizontal table and the other end has a mass of 5 kg attached to it. The 5 kg mass revolves uniformly on the table about the point P . Determine the maximum number of complete revolutions the mass can make in a minute without breaking the string. (Use $g = 9.8 \text{ m/s}^2$.) **3**

Question 14 (15 marks)

- (a) The region bounded by $y = x^4 + 1$, the x -axis, the y -axis and the line $x = 1$ is rotated about the line $x = 3$. Use the method of cylindrical shells to find the volume of the solid generated. 4

- (b) z is a point in the first quadrant of the Argand diagram which lies on the circle $|z - 3| = 3$. Given $\arg(z) = \theta$, find $\arg(z^2 - 9z + 18)$ in terms of θ . 2

- (c) The point $P(x_1, y_1)$ lies on the ellipse $\frac{x^2}{10} + \frac{y^2}{16} = 1$ where $x_1 \neq 0$ and $y_1 \neq 0$.



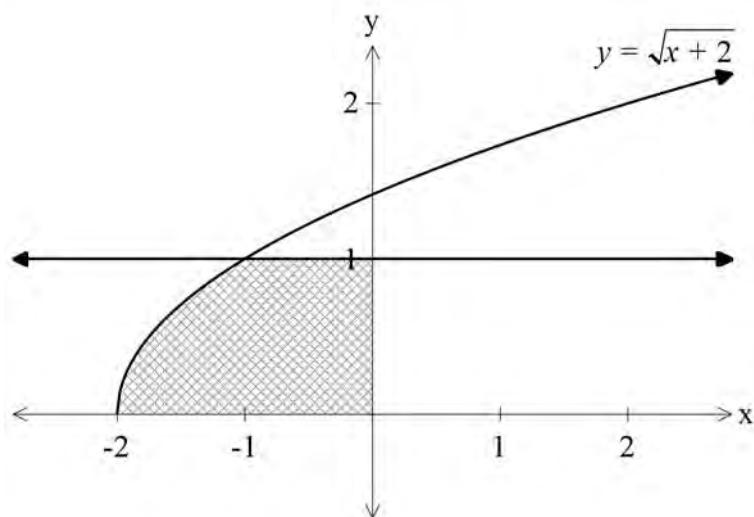
- (i) Show that the equation of the tangent at P is $\frac{xx_1}{10} + \frac{yy_1}{16} = 1$. 2
- (ii) The tangent at P cuts the y -axis at T . Determine the coordinates of the point T . 1
- (iii) Find the coordinates of the foci, S and S' . 2
- (iv) Find the equations of the directrices. 1
- (v) Show that $\frac{PS}{PS'} = \frac{TS}{TS'}$. 3

Question 15 (15 marks)

(a) Evaluate $\int_1^9 e^{\sqrt{x}} dx$.

4

- (b) The base of a solid is the area (as shaded below) enclosed by the curve $y = \sqrt{x+2}$, the x -axis, the y -axis and the line $y=1$.



Each cross-section of the solid perpendicular to the x -axis is a regular hexagon with one side in the base of the solid. Find the volume of the solid.

4

- (c) The polynomial $P(x) = 2x^3 - 4x^2 + 7x + 5$ has roots α , β and γ . Find the value of :

(i) $\alpha^3 + \beta^3 + \gamma^3$

3

(ii) $\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta$

2

- (d) a and b are real non-zero numbers and $\omega = \frac{az+b}{bz+a}$. Given $|z|=1$, show that $|\omega|=1$.

2

Question 16 (15 marks)

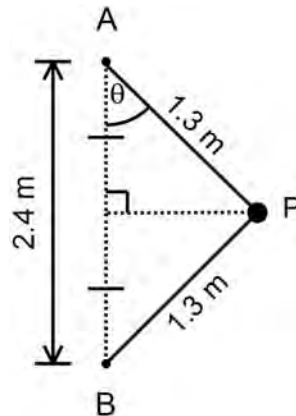
- (a) The equation $px^3 + qx + r = 0$ has roots α , β and γ . Find the 4

polynomial equation with roots $\frac{1}{-\alpha\beta - \alpha\gamma}$, $\frac{1}{-\alpha\beta - \beta\gamma}$ and $\frac{1}{-\alpha\gamma - \beta\gamma}$.

Express your answer in simplest form.

- (b) A particle of mass m kg, is attached at P by two strings, each of length 1.3 m, to two fixed points, A and B , which are 2.4 m apart and lie on a vertical line, as shown in the diagram below.

The particle moves with constant speed v ms⁻¹ in a horizontal circle about the midpoint of AB so that both pieces of string experience tension. Let T_1 and T_2 represent the tension in AP and BP respectively. The acceleration due to gravity is g ms⁻².

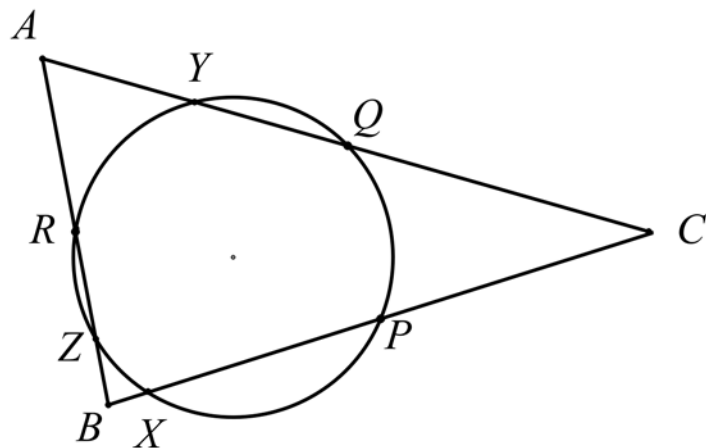


- (i) Draw a diagram showing all the forces acting on P . 1
- (ii) Resolve the forces on P in both the horizontal and vertical directions. 2
- (iii) Find the tension in each part of the string in terms of m , v and g . 2
- (iv) Show that $v > \frac{\sqrt{30g}}{12}$. 2

Question 16 continues on the next page

Question 16 (continued)

- (c) P , Q and R are the midpoints of BC , AC and AB respectively. The circle drawn through P , Q and R intersects BC , AC and AB a second time at X , Y and Z respectively as shown below.



- | | | |
|-------|--|----------|
| (i) | Explain why $RPCQ$ is a parallelogram. | 1 |
| (ii) | Show $\triangle XQC$ is isosceles. | 2 |
| (iii) | Show $AX \perp BC$. | 1 |

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



Sydney Girls High School

Mathematics Faculty

Multiple Choice Answer Sheet

2014 Trial HSC Mathematics Extension 2

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☒ D ☐
2. A ☐ B ☒ C ☐ D ☐
3. A ☐ B ☒ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☒ C ☐ D ☐
6. A ☐ B ☒ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☒
8. A ☐ B ☐ C ☒ D ☐
9. A ☐ B ☒ C ☐ D ☐
10. A ☒ B ☐ C ☐ D ☐

QUESTION 11

Comments

$$(a)(i) \quad \frac{13}{(x+2)(x^2+9)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+9}$$

$$13 = A(x^2+9) + (Bx+C)(x+2)$$

let $x = -2$ $13 = 13A$ $A = 1$

coeff. of x^2 $0 = A + B$ $B = -A = -1$

constant $13 = 9A + 2C$ $C = \frac{13-9}{2} = 2$

$$(ii) \quad \int \frac{13 dx}{(x+2)(x^2+9)} = \int \left(\frac{1}{x+2} + \frac{-x+2}{x^2+9} \right) dx$$

$$= \ln(x+2) - \frac{1}{2} \ln(x^2+9) + \frac{2}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

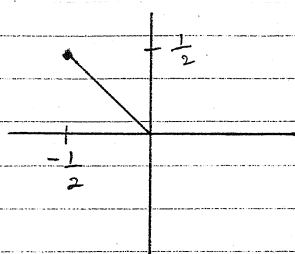
(a)(ii)

Common mistake was a minus sign in front of $\frac{2}{3} \tan^{-1}\left(\frac{x}{3}\right)$

$$(b)(i) \quad z = \frac{1+2i}{1-3i} \times \frac{1+3i}{1+3i}$$

$$= \frac{1+2i+3i-6}{1+9}$$

$$= -\frac{1}{2} + \frac{i}{2}$$



$$|z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{\sqrt{2}}$$

$$\arg(z) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\therefore z = \frac{1}{\sqrt{2}} \operatorname{cis} \frac{3\pi}{4}$$

$$(ii) \quad z^7 = \left(\frac{1}{\sqrt{2}} \operatorname{cis} \frac{3\pi}{4} \right)^7$$

$$= \frac{1}{8\sqrt{2}} \operatorname{cis} \frac{21\pi}{4}$$

$$= \frac{1}{8\sqrt{2}} \operatorname{cis} \left(-\frac{3\pi}{4} \right)$$

$$= \frac{1}{8\sqrt{2}} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right)$$

$$= -\frac{1}{16} - \frac{1}{16}i$$

(b)(i)

A number of students calculate $\arg(z)$ incorrectly as $-\frac{\pi}{4}$. A diagram would have eliminated this mistake.

$$(c)(i) \quad (4+3i)^2 = 16 + 24i + 9i^2 = 7 + 24i$$

$$(ii) \quad z = i \pm \sqrt{i^2 - 4 \times 2(-1-3i)}$$

$$= i \pm \sqrt{-1 + 8 + 24i}$$

$$= i \pm \sqrt{7 + 24i}$$

$$= i \pm \frac{4+3i}{4} \quad \text{using (i)}$$

$$= \frac{i+4+3i}{4} \quad \text{or} \quad \frac{i-4-3i}{4}$$

i.e. $z = 1+i \quad \text{or} \quad -1-\frac{1}{2}i$

(c)(ii)

Some students did not think to use the quadratic formula and some did not make the link to the result in (i).

Question 11 (continued)

Comments

(d) Since coeff. of $P(x)$ are real and $1+3i$ is a root, then $1-3i$ is also a root.

Let roots be $\alpha, \beta, 1 \pm 3i$

Sum of roots: $\alpha + \beta + 1 + 3i + 1 - 3i = -2 \quad \therefore \alpha + \beta = -4$

Product of roots: $\alpha \beta (1+3i)(1-3i) = 90$

$\alpha \beta (1+9) = 90 \quad \therefore \alpha \beta = 9$

$\therefore$ Eqn. with roots α and β is

$$z^2 + 4z + 9 = 0$$

$$(z+2)^2 = -5$$

$$z = -2 \pm \sqrt{-5}$$

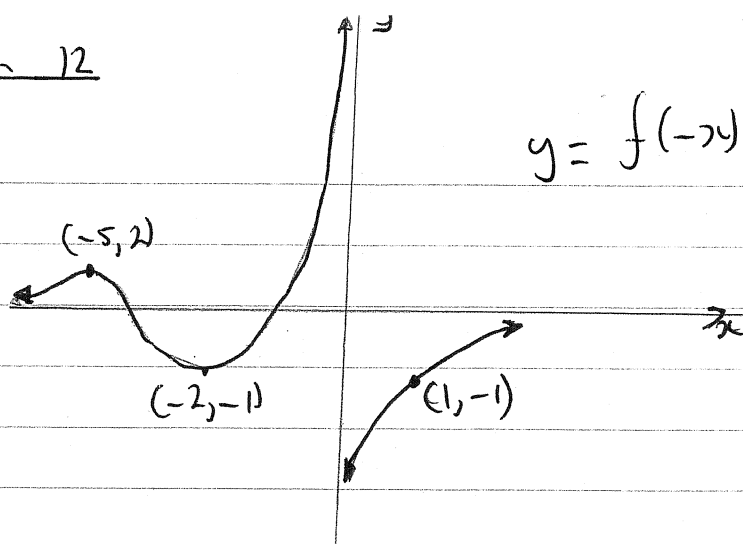
$\therefore$ The roots of $P(x)$ are $1+3i, 1-3i, -2+\sqrt{5}i, -2-\sqrt{5}i$.

(d) Some solutions were too long-winded.

Question 12

a) i)

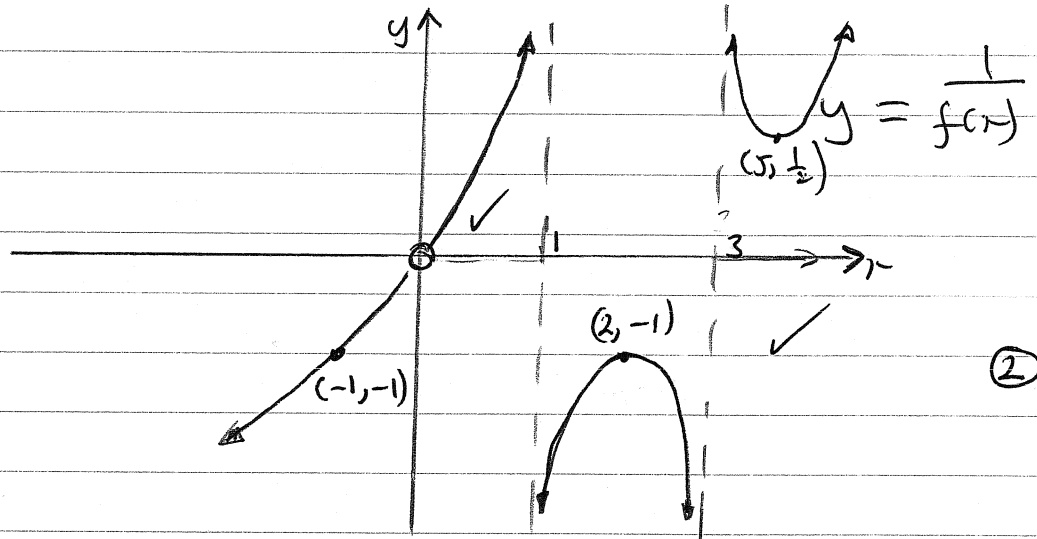
$$y = f(-x)$$



①

①
Correct
Answer.

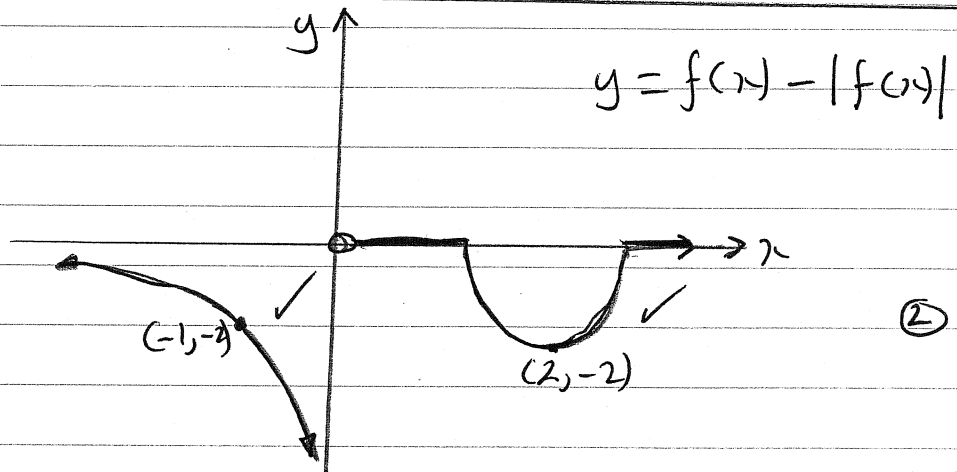
ii)



②

②

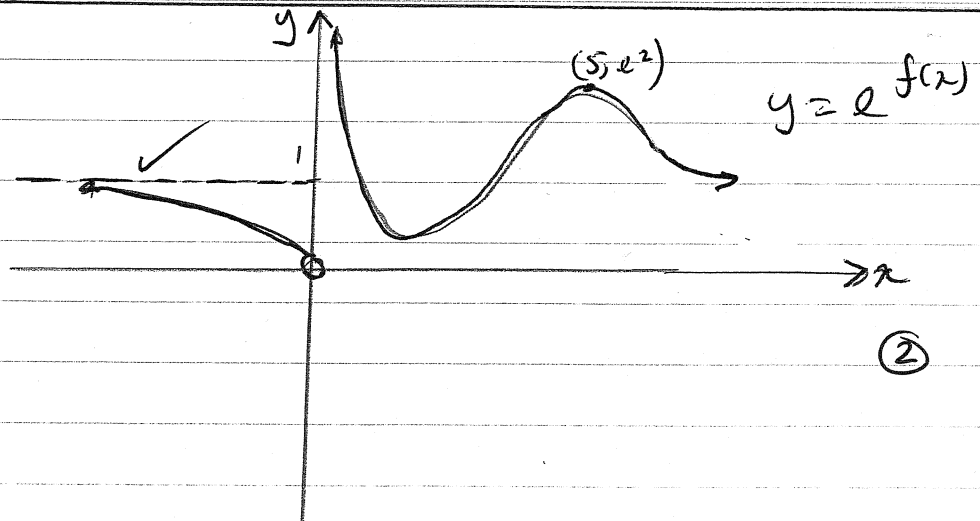
iii)



②

②

iv)



②

②

$$b) \quad x^2 - 2x + 10 = (x-1)^2 + 3^2$$

(3)

$$I = \int \frac{2x-2}{\sqrt{(x-1)^2 + 3^2}} + \int \frac{3 dx}{\sqrt{(x-1)^2 + 3^2}} \checkmark$$

$$= I_1 + I_2$$

$$\text{Let } u = x^2 - 2x + 10 \quad \left| \quad = 3 \log_e [(x-1) + \sqrt{x^2 - 2x + 10}] + C$$

$$du = (2x-2) dx$$

$$I_1 = \int \frac{du}{u^{\frac{1}{2}}}$$

$$= 2 u^{\frac{1}{2}}$$

$$= 2 \sqrt{x^2 - 2x + 10} \checkmark$$

$$\therefore I = 2\sqrt{x^2 - 2x + 10} + 3 \log_e [(x-1) + \sqrt{x^2 - 2x + 10}] + C \checkmark$$

$$c) \quad i) m = \left(-\frac{2}{p} + \frac{2}{q} \right) \div (2p - 2q) \checkmark$$

$$= \frac{1}{pq}$$

Eqn p & q

(2)

$$y + \frac{2}{p} = \frac{1}{pq} (x - 2p)$$

$$pqy + 2q = x - 2p$$

$$x - pqy = 2p + 2q \checkmark$$

$$ii) \quad \text{sub } x=0, y=-6 \text{ in above}$$

$$0 + 6pq = 2(p+q)$$

$$3pq = p+q \checkmark \quad (1)$$

Co-ords of midpoint

$$x = \frac{2p+2q}{2}$$

$$y = \left(-\frac{2}{p} - \frac{2}{q} \right) \div 2$$

$$x = p+q \checkmark$$

$$y = -\frac{p+q}{pq} \checkmark$$

(3)

sub (1) in above

$$y = -\frac{3pq}{pq}$$

$$y = -3 \checkmark$$

13 (a) (i) let $u = (1-x^2)^{\frac{n}{2}}$ $u' = 1$

$$u' = \frac{n}{2} (1-x^2)^{\frac{n}{2}-1} \cdot x \cdot 2x \quad v = x$$

$$I_n = \left[+x (1-x^2)^{\frac{n}{2}} \right]_0^1 + \frac{n}{2} \int x^2 (1-x^2)^{\frac{n-2}{2}} dx - n \int (1-x^2)^{\frac{n-2}{2}} dx + n I_{n-2}$$

$$= 0 + n \int (x^2-1)(1-x^2)^{\frac{n-2}{2}} dx + n I_{n-2}$$

$$= -n I_n + n I_{n-2} \quad \checkmark \checkmark \checkmark$$

$$(1+n)I_n = n I_{n-2}$$

$$I_n = \frac{n}{n+1} I_{n-2} \quad \text{G.E.O.}$$

(ii) $I_5 = \frac{5}{2} I_3$

$$= \frac{5}{2} \times \frac{3}{2} I_1$$

$$= \frac{2}{3} \times \frac{\pi}{4}$$

$$= \frac{5\pi}{32} \quad \checkmark \checkmark$$

$$I = \int_0^1 (1-x^2)^{\frac{1}{2}} dx$$

$$= \frac{\pi \times 1^2}{4}$$

$$= \frac{\pi}{4}$$



This is easier than using $u = \sin x$.

(b) (i) $(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$

$$= \cos^5 \theta + i \sin^5 \theta$$

$$\cos^5 \theta = \cos^5 \theta - 10 \cos^3 \theta (1-\cos^2 \theta) + 5 \cos \theta (1-\cos^2 \theta)^2$$

$$= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta$$

$$= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad \checkmark \checkmark \checkmark$$

(ii) let $\cos 5\theta = 0$

$$\cos \theta (16 \cos^4 \theta - 20 \cos^2 \theta + 5) = 0$$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

$$\therefore x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10} \quad \checkmark \checkmark \checkmark$$

(c) $T \leq 12 \times 9.8$

$$\leq 117.6 \text{ N}$$

$$mr\omega^2 \leq 117.6$$

$$5 \times 0.4 \times \omega^2 \leq 117.6$$

$$\omega^2 \leq 58.8$$

$$\omega \leq \sqrt{58.8}$$

$$\text{max rev} = \frac{\sqrt{58.8}}{2\pi} \times 60$$

$$= 73.22511207$$

$$\approx 73$$

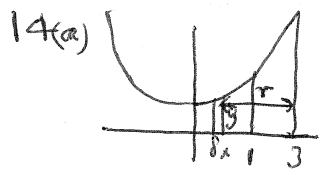
The answer

requires the no.

of rev in 1 min

not 1 sec.

$\checkmark \checkmark \checkmark$



Students had more success using 1st principles rather than formulae.

$$V_{\text{shell}} = \pi \{ (3-x)^2 - (3-x-x)^2 \}$$

$$= \pi \{ (6-2x)x \}$$

$$V_{\text{solid}} = 2\pi \int_0^1 (3-x)y \, dx$$

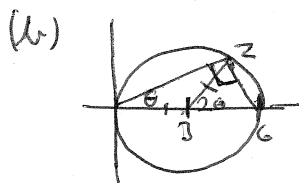
$$= 2\pi \int_0^1 (3-x)(x^2+1) \, dx$$

$$= 2\pi \int_0^1 (3x^2 + 3 - x^3 - x) \, dx$$

$$= 2\pi \left[\frac{3x^3}{3} + 3x - \frac{x^4}{4} - \frac{x^2}{2} \right]_0^1$$

$$= 2\pi \times \left(\frac{3}{3} + 3 - \frac{1}{4} - \frac{1}{2} \right)$$

$$= \frac{88\pi}{15} \quad \checkmark \checkmark \checkmark$$



The geometric approach was much easier than the algebraic.

$$\arg(z^2 - 9z + 18)$$

$$= \arg\{(z-6)(z-3)\}$$

$$= \arg(z-6) + \arg(z-3)$$

$$= \theta + \frac{\pi}{2} + 2\theta$$

$$= 3\theta + \frac{\pi}{2} \quad \checkmark \checkmark$$

$$(c)(i) \quad \frac{2x}{10} + \frac{2y}{16} \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{5} \times \frac{8}{5}$$

$$= -\frac{8x}{5}$$

$$m_t = \frac{8x_1}{5y_1}$$

$$y - y_1 = \frac{8x_1}{5y_1} (x - x_1)$$

$$5y_1 y - 5y_1^2 = 8x_1 x - 8x_1^2$$

$$8x_1 x + 5y_1 y = 8x_1^2 + 5y_1^2$$

$$\frac{x_1 x}{10} + \frac{y_1 y}{16} = \frac{x_1^2}{10} + \frac{y_1^2}{16} = 1 \quad \text{Q.E.D.}$$

$$(ii) \quad \frac{yy_1}{16} = 1$$

$$x=0, \quad y = \frac{16}{y_1} \quad \checkmark$$

$$(iii) \quad 10 = 16(1-e^2)$$

$$\frac{5}{8} = 1-e^2$$

$$e^2 = \frac{3}{8}$$

$$e = \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{\sqrt{6}}{4}$$

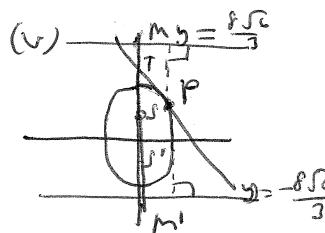
$$\text{foci are } (0, \pm \frac{\sqrt{6}}{4} \times 4)$$

$$(0, \pm \sqrt{6}) \quad \checkmark \checkmark \checkmark$$

$$(iv) \quad y = \pm \frac{4}{\frac{\sqrt{2}}{4}}$$

$$= \pm \frac{16}{\sqrt{2}}$$

$$= \pm \frac{8\sqrt{2}}{3} \quad \checkmark$$



$$\frac{PS}{PS'} = \frac{ePM}{ePM'}$$

This is easier than trying to find PS or PS'.

$$= \frac{\sqrt{\frac{8\sqrt{2}}{3} - y_1}}{\sqrt{y_1 - \frac{8\sqrt{2}}{3}}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{16 - \sqrt{2}y_1}{16 + \sqrt{2}y_1}$$

$$\frac{TS}{TS'} = \frac{\frac{16}{y_1} - \sqrt{2}}{\frac{16}{y_1} + \sqrt{2}}$$

$$= \frac{16 - \sqrt{2}y_1}{16 + \sqrt{2}y_1}$$

Q.E.D. $\checkmark \checkmark \checkmark$

Question 15

a) $I = \int_1^9 \frac{x}{\sqrt{x}} dx$
 let $u = \sqrt{x}$
 $x = u^2$

when $x = 9$, $u = 3$
 $x = 1$, $u = 1$ ✓

$dx = 2u du$
 $I = 2 \int u e^u du$ ✓

$= 2 \left[u e^u - \int e^u \right] du$ [IBP+]
 $= 2 \left[u e^u - e^u \right]_1^3$
 $= 2 \left[(3e^3 - e^3) - (e^1 - e^1) \right]$ ✓
 $= 4e^3$ ✓

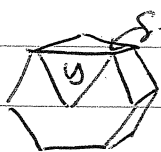
(4)

b) Volume for $x=0$ to $x=-1$ is a hexagonal prism



$V_1 = 6 \times \frac{1}{2} \times 1 \times 1 \times \sin 60^\circ \times 1$
 $= 3 \times \frac{\sqrt{3}}{2}$
 $= \frac{3\sqrt{3}}{2} \text{ units}^3$ ✓

Volume for $x=-1$ to $x=-2$



$V_2 = \frac{3\sqrt{3}}{2} \int_{-2}^{-1} (x+2) dx$ ✓

$= \frac{3\sqrt{3}}{2} \left[\frac{x^2}{2} + 2x \right]_{-2}^{-1}$ ✓

$V_{\text{olia}} = 6 \times \frac{1}{2} y^2 \sin 60^\circ$
 $= \frac{3\sqrt{3}}{2} y^2 \sin 60^\circ$

$= \frac{3\sqrt{3}}{2} \left[\left(\frac{1}{2} - 2 \right) - \left(2 - 4 \right) \right]$
 $= \frac{3\sqrt{3}}{4}$ ✓

(4)

$\therefore \text{Total Volume} = \frac{3\sqrt{3}}{2} + \frac{3\sqrt{3}}{4}$
 $= \frac{9\sqrt{3}}{4} \text{ units}^3$

Q15

$$1) 2x^3 = 4x^2 - 7x - 5$$

$$x^3 = 2x^2 - \frac{7x}{2} - \frac{5}{2}$$

$$\alpha^3 = 2\alpha^2 - \frac{7\alpha}{2} - \frac{5}{2}$$

$$\beta^3 = 2\beta^2 - \frac{7\beta}{2} - \frac{5}{2} \quad \checkmark$$

$$\gamma^3 = 2\gamma^2 - \frac{7\gamma}{2} - \frac{5}{2}$$

$$\alpha^3 + \beta^3 + \gamma^3 = 2(\alpha^2 + \beta^2 + \gamma^2) - \frac{7}{2}(\alpha + \beta + \gamma) - 3\left(\frac{5}{2}\right)$$

$$= 2(-3) - \frac{7}{2}(2) - \frac{5}{2}(3)$$

$$= -\frac{41}{2} \quad \checkmark$$

$$(\alpha^2 + \beta^2 + \gamma^2)$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= (2)^2 - (-2)\left(\frac{7}{2}\right)$$

$$= -3 \quad \checkmark$$

(3)

$$ii) \alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta \quad \text{or using}$$

$$\alpha^2(\beta + \gamma) + \beta^2(\alpha + \gamma) + \gamma^2(\alpha + \beta)$$

$$\alpha + \beta + \gamma = 2$$

$$\alpha^2(2 - \alpha) + \beta^2(2 - \beta) + \gamma^2(2 - \gamma)$$

$$= 2\alpha^2 - \alpha^3 + 2\beta^2 - \beta^3 + 2\gamma^2 - \gamma^3$$

$$= 2(\alpha^2 + \beta^2 + \gamma^2) - (\alpha^3 + \beta^3 + \gamma^3)$$

$$= 2(-3) - \left(-\frac{41}{2}\right)$$

$$= \frac{29}{2}$$

$$(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$- 3(\alpha\beta\gamma)$$

$$= 2\left(\frac{7}{2}\right) - 3\left(-\frac{5}{2}\right)$$

$$= 7 + \frac{15}{2}$$

$$= \frac{29}{2}$$

(2)

Correct soln

$$d) w = \frac{az+b}{bz+a} \quad \text{①}$$

$$|z| = 1 \Rightarrow z = \cos\theta + i\sin\theta \quad \text{②}$$

$$w = \frac{a\cos\theta + b + ai\sin\theta}{b\cos\theta + a + bi\sin\theta}$$

$$|w| = \frac{|(a\cos\theta + b) + ai\sin\theta|}{|(b\cos\theta + a) + bi\sin\theta|}$$

$$= \frac{\sqrt{(a\cos\theta + b)^2 + (a\sin\theta)^2}}{\sqrt{(b\cos\theta + a)^2 + (b\sin\theta)^2}}$$

$$= \frac{\sqrt{a^2\cos^2\theta + 2ab\cos\theta + b^2 + a^2\sin^2\theta}}{\sqrt{b^2\cos^2\theta + 2ab\cos\theta + a^2 + b^2\sin^2\theta}}$$

$$= \frac{\sqrt{a^2 + b^2 + 2ab\cos\theta}}{\sqrt{a^2 + b^2 + 2ab\cos\theta}}$$

$$= 1$$

$$|w| = 1$$

$$\text{or } w = \frac{az+b}{bz+a}$$

$$wbz + wa = az + b$$

$$wbz - az = b - wa$$

$$z(wb - a) = b - wa$$

$$|z| = \frac{|b - wa|}{|wb - a|} \text{ and } |z| = 1$$

$$\therefore b - wa = wb - a$$

$$b + a = w(b + a)$$

$$w = \frac{b+a}{b+a}$$

$$= 1$$

Correct Soln (2)

QUESTION 16

Comments

(a)

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{q}{p}$$

$$\alpha\beta\gamma = -\frac{r}{p}$$

$$x = \alpha, \beta, \gamma$$

New eqn. has roots $y = \frac{1}{-\alpha\beta - \alpha\gamma}, \frac{1}{-\alpha\beta - \beta\gamma}, \frac{1}{-\alpha\gamma - \beta\gamma}$

$$= \frac{1}{-\alpha(\beta + \gamma)}, \frac{1}{-\beta(\alpha + \gamma)}, \frac{1}{-\gamma(\alpha + \beta)}$$

$$= \frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$$

$$\therefore y = \frac{1}{x^2} \Rightarrow x = \frac{1}{\sqrt{y}}$$

$$p\left(\frac{1}{\sqrt{y}}\right)^3 + q\left(\frac{1}{\sqrt{y}}\right) + r = 0$$

$$\frac{1}{\sqrt{y}}\left(\frac{p}{y} + q\right) = -r$$

$$\frac{1}{y}\left(\frac{p^2}{y^2} + \frac{2pq}{y} + q^2\right) = r^2$$

$$p^2 + 2pqy + q^2y^2 = r^2y^3$$

$$\therefore r^2y^3 - q^2y^2 - 2pqy - p^2 = 0$$

The solution is one of two common approaches to this question. The other approach used the relationship

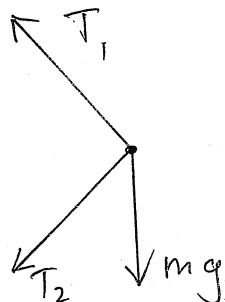
$$-\alpha\beta - \alpha\gamma = \beta\gamma - \frac{q}{p}$$

as well as

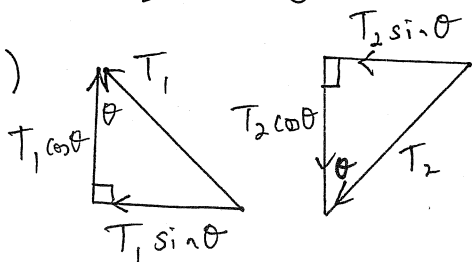
$$\beta\gamma = \frac{-r}{p\alpha}$$

Many students made small errors and many did not make a start - writing the sum, product etc would have guaranteed the 1st mark.

(b) (i)



(ii)



Horizontally :

$$T_1 \sin \theta + T_2 \sin \theta = \frac{mv^2}{r}$$

Vertically :

$$T_1 \cos \theta - T_2 \cos \theta - mg = 0$$

(b)(i) Marked generously - but the three vectors shown in the solution was expected (No more, no less!)

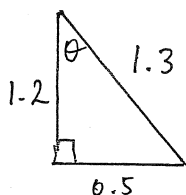
(ii) Take care with the position of angle θ .

Question 16 (continued)

Comments

(b) continued

$$\begin{aligned} \text{(iii)} \quad & \left. \begin{aligned} \sin \theta (T_1 + T_2) &= \frac{mv^2}{r} \\ \cos \theta (T_1 - T_2) &= mg \end{aligned} \right\} \end{aligned}$$



$$\begin{aligned} \sin \theta &= \frac{5}{13} \\ \cos \theta &= \frac{12}{13} \\ r &= 0.5 \end{aligned}$$

$$T_1 + T_2 = \frac{13}{5} \left(\frac{mv^2}{0.5} \right) = \frac{26mv^2}{5}$$

$$T_1 - T_2 = \frac{13mg}{12}$$

$$2T_1 = \frac{26mv^2}{5} + \frac{13mg}{12}$$

$$\text{i.e. } T_1 = \frac{13mv^2}{5} + \frac{13mg}{24}$$

$$2T_2 = \frac{26mv^2}{5} - \frac{13mg}{12}$$

$$T_2 = \frac{13mv^2}{5} - \frac{13mg}{24}$$

(iv) $T_2 > 0$ (since both strings experience tension)

$$\frac{13mv^2}{5} - \frac{13mg}{24} > 0 \quad \therefore \frac{v^2}{5} - \frac{g}{24} > 0 \quad v^2 > \frac{5g}{24}$$

$$v > \sqrt{\frac{5g}{24}} \quad v > \sqrt{\frac{30g}{144}} \quad \text{i.e. } v > \frac{\sqrt{30g}}{12}$$

(b)(iii)

You must eliminate θ from your answers.

You do not need to find θ - you should use the triangle to determine $\sin \theta$ and $\cos \theta$.

(ii) You had to show the given relationship. Some solutions were left incomplete.

(c) (i) $\frac{AQ}{QC} = \frac{AR}{RB} = 1$ (Q and R are midpts of AC and AB)

$\therefore RQ \parallel BC$ (ratio of intercepts is equal on parallel lines)

Similarly $\frac{BP}{PC} = \frac{BR}{RA} = 1$ and $PR \parallel AC$.

$\therefore RPCQ$ is a parallelogram (2 pairs of opp. sides parallel)

(ii) $\angle QCP = \angle QRP$ (opp. $\angle$ s of parallelogram)

$\angle QRP = \angle QXP$ ($\angle$ s in same segment)

$\therefore \angle QCP = \angle QXP \quad \therefore \triangle XQC$ is isosceles (2 equal $\angle$ s)

(iii) $AQ = QC$ (given, Q is mid-point of AC)

$QX = QC$ (sides opp. equal $\angle$ s in isos. $\triangle$)

$\therefore$ Circle passes through A, X, C with centre Q. ($AQ = QC = QX$)

$\therefore \angle AXC = 90^\circ$ ($\angle$ in semi-circle) $\therefore AX \perp BC$

(c)(i)

Poorly done in general - many "solutions" presented insufficient data.

(ii) This was not a difficult question if you use the result from (i).

(iii) More detail needed from most attempts.